



Indian Space Olympiad (ISO) 2026

SYLLABUS -INTERMEDIATE LEVEL (CLASSES VII–VIII)

01. Fundamentals of Astronomy

History of astronomy, celestial sphere, celestial coordinates (basic), astronomical observations, telescopes, electromagnetic spectrum (introductory), astronomical measurements.

02. The Solar System

Solar Nebula Theory, the Sun, planetary motion, terrestrial and giant planets, dwarf planets, planetary atmospheres, asteroids, comets, Kuiper Belt, Near-Earth Objects.

03. Stars, Galaxies & the Universe

Life cycle of stars, stellar classification (basic), nebulae, star clusters, galaxies, Milky Way, black holes (introductory), origin of the Universe.

04. Earth & Moon

Earth's interior, atmosphere, magnetosphere, climate, lunar surface, Moon formation, eclipses, tides, Earth-Moon system.

05. Satellites & Space Technology

Satellite orbits, communication systems, navigation, Earth observation, remote sensing, spacecraft components.

06. Rockets & Launch Vehicles

Rocket propulsion, escape velocity (concept), launch vehicle stages, satellite launches, reusable launch technology.

07. ISRO & India's Space Missions

PSLV, GSLV, LVM3, Chandrayaan series, Mars Orbiter Mission, Aditya-L1, Gaganyaan, XPoSat, SpaDeX, future Indian missions.

08. Space Exploration

Human spaceflight, International Space Station, lunar exploration, Mars exploration, planetary missions, space telescopes.

09. Basic Physics for Space Science

Newton's Laws of Motion, gravitation, circular motion, work, energy, momentum, pressure, heat, optics, electricity and magnetism.

10. Scientific Reasoning & Logical Aptitude

Scientific investigations, experimental analysis, graph interpretation, logical reasoning, quantitative aptitude, problem-solving.

11. General Space Awareness

Recent ISRO missions, international space missions, astronomy discoveries, space technology, emerging trends in space science.